Zejia Lin 林泽佳

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Keywords: Inference Optimization, Compiler, Mixed-precision

- Strong background on computer architecture, system and software. Focusing GPU software designs (compiler/runtime) around resource sharing and task scheduling.
- Interend at Tencent. Published 3 paper on GPUs. Familiar with CUDA, LLVM, SGLang, veRL.

Education

Sun Yat-sen University, Guangzhou, China — *2022 – 2027 (expected)*
Ph.D. student of Computer Science — Advisors: Xianwei Zhang, Yutong Lu

Northwestern Polytechnical University, Xi'an, China — *2018 – 2022*
Bachelor Degree of Software Engineering — Rank: 22 / 259 — Outstanding Graduate

Publications

- **Zejia Lin**, Hongxin Xu, Guanyi Chen, Zhiguang Chen, Yutong Lu and Xianwei Zhang
  [ASPLOS'26, CCF-A] Bullet: Boosting GPU Utilization for LLM Serving via Dynamic Spatial-Temporal Orchestration. *The 31st ACM International Conference on Architectural Support for Programming Languages and Operating Systems, 2026.*
- **Zejia Lin**, Aoyuan Sun, Xianwei Zhang and Yutong Lu
  [LCTES'24, CCF-B] MixPert: Optimizing Mixed-Precision Floating-Point Emulation on GPU Integer Tensor Cores. *The 25th ACM SIGPLAN/SIGBED International Conference on Languages, Compilers, and Tools for Embedded Systems, 2024.*
- **Zejia Lin**, Zewei Mo, Xuanteng Huang, Xianwei Zhang and Yutong Lu
  [ICCD'23, CCF-B] KeSCo: Compiler-based Kernel Scheduling for Multi-task GPU Applications. *The IEEE 41th International Conference on Computer Design, 2023.*
- Wenxuan Pan, **Zejia Lin**, Jiangsu Du and Xianwei Zhang
  [TACO, CCF-A] HuntKTm: Hybrid Scheduling and Automatic Management for Efficient Kernel Execution on Modern GPUs. *ACM Transactions on Architecture and Code Optimization, 2025.*
- Kan Wu, **Zejia Lin**, Mengyue Xi, Zhongchun Zheng, Wenxuan Pan, Xianwei Zhang and Yutong Lu
  [DAC'25, CCF-A] GoPTX: Fine-grained GPU Kernel Fusion by PTX-level Instruction Weaving. *The 62nd ACM/IEEE Design Automation Conference, 2025.*

Experiences

Research Intern, Wechat Search, Tencent — *2024.12 – Now*

Research on improving LLM inference performance in RAG scenarios. Proposed Bullet, a method enabling single-GPU PD-disaggregation via temporal-spatial sharing, deployed online and validated through mirrored traffic tests to outperform existing approaches in both latency and throughput. This work has been accepted by ASPLOS' 26 (acceptance rate: 10.6%) and presented at the SGLang Beijing Meetup.

Teaching Assistant, Compiler Principles, Sun Yat-sen University — *2023.03 – 2023.07*

Teaching assistant of undergraduate course, implement a compiler for a subset of C language. Introduce LLVM framework in class and release programming homeworks.

Backend Develop Intern, Wechat Pay, Tencent — *2021.07 – 2021.09*

Build a Django-based Wechat robot to manage backend testing issues from develop team. Add features to access Tencent's TAPD project management system.

Skills

- Programming Languges: C/C++, CUDA, Python
- Compiler Toolchain: LLVM, CMake, Clang, GDB
- GPU Tools: NVIDIA nsys/ncu, CUTLASS, cuBLAS
- AI Infra: PyTorch, vLLM, SGLang, veRL
- CCF CSP Certification: Top 5.2%
- English CET-6: 610 / 750

Honors & Awards

Sun Yat-sen University *First Class Scholarship*	*Year 2022 – 2024*
Tencent *Excellence Scholarship*	*Year 2023*
Northwestern Polytechnical University *Distinguished Graduate*	*Year 2022*
Northwestern Polytechnical University *First Class Scholarship*	*Year 2019 – 2021*
CCCC Wechat Mini Program Development *Second Price (Leader)*	*2022.08*

Profesional Service

- ACM EuroSys 2026, Artifact Evaluation Committe
- ACM SOSP 2025, Artifact Evaluation Committe
- IEEE NAS 2024, Sub-reviewer
- IEEE ICPADS 2022, Sub-reviewer